FLAGSHIP PROJECT 02 · EXECUTIVE LAYER

National Digital Health Investment Strategy

Executive Summary

Kenya's digital health ambitions — an interoperable national health information system, telemedicine at scale, AI-assisted care, remote monitoring, and rehabilitation delivered at distance — all compete for the same constrained fiscal space that is already stretched by the SHA transition and a chronic health-workforce funding gap. This brief evaluates four investment areas on current readiness and expected impact, to recommend a sequence rather than a wish list.

1. Where Kenya Stands Today

Kenya is rated Phase 2 (of a higher-numbered maturity scale) on the Global Digital Health Monitor as of 2023, with low interoperability maturity across leadership, governance, workforce and technology domains. The structural problem is fragmentation, not absence of ambition: the Ministry of Health, all 47 county governments, and the Social Health Authority each operate separate health information systems, budget-tracking tools and claims platforms — 49 purchasing bodies with no single data backbone connecting them.

The Digital Health Act (2023) responds to this directly. It establishes a Digital Health Agency mandated to build a Comprehensive Integrated Health Information System, and the accompanying Digital Health Superhighway blueprint defines National Shared Registries and an Enterprise Service Bus as the technical spine for interoperability. The policy architecture is sound; what is missing is completed execution.

Sources: PMC12165431 Homa Bay digital health landscape assessment; Amref Health Africa, Oct 2024; KELIN Kenya, Digital Health Bill analysis; PMC13449493 KenyaEMR maturity assessment.

2. Area-by-Area Assessment

EHR & Interoperability — Foundational, in progress.

KenyaEMR and related electronic medical record systems have matured over 15 years of policy iteration (2009 HIS Policy through the 2023 Digital Health Act), but interoperability between systems — public-private, county-national, and across the 49 purchasing bodies — remains the binding constraint on every other digital health investment. Every other area in this brief depends on this layer working.

Telemedicine — Proven in pilots, constrained by infrastructure.

Hybrid telehealth-plus-Community-Health-Worker models have demonstrated real effectiveness in Kenya — a Teso North maternal and child health pilot linking virtual consults to CHWs improved postnatal care timeliness. The Ministry of Health is developing dedicated Telemedicine Policy and Guidelines. The main technical barrier is connectivity: a 2025 architecture study proposes edge computing and lightweight on-device AI specifically to keep telemedicine functional in Kenya's low-bandwidth, intermittently connected rural areas.

Remote Patient Monitoring & AI — Early-stage, real interest, thin regulation.

AI is actively being discussed, piloted and commercialised in Kenyan healthcare, especially inside telemedicine and mobile health products, but dedicated AI-in-health policy is still emerging rather than settled. Remote monitoring (wearables, SMS-based platforms such as PROMPTS for maternal health) has documented reach in Kenya; broader device-based monitoring evidence is currently stronger at the regional African level than Kenya-specific, and should be read with that caveat.

Tele-rehabilitation — Real unmet need, almost no Kenya-specific evidence yet.

This is the most honest finding in this brief: tele-rehabilitation has a credible evidence base across rural Africa broadly (a 2024 systematic review found meaningful gains in functional recovery and quality of life), and Kenya's rehabilitation workforce shortage makes the case for it strong on paper. But a targeted search turned up essentially no Kenya-specific tele-rehabilitation studies or pilots — this is a genuine white space, not a mature investment area. It should be funded as an evaluated pilot generating local evidence, not assumed to work because it works elsewhere.

Sources: Frontiers in Digital Health, MNCH telemedicine hybrid model, Dec 2025; World Journal of Public Health, edge-based telemedical architecture for Kenya, 2025; CIPIT Strathmore, AI in Health policy pathways, Aug 2025; Springer Bulletin of Faculty of Physical Therapy, tele-rehabilitation in African rural areas systematic review, 2024; ClinicalTrials.gov NCT05110521, PROMPTS digital health package, Kenya.

3. Investment Sequencing

Readiness varies sharply across the four areas — and each later area depends on the ones before it working. Sequencing, not simultaneous investment, is the realistic path.

Investment Area	Readiness Today	Recommended Horizon
EHR & Interoperability (Digital Health Superhighway)	Policy exists, execution early — foundational	Invest now — sequencing priority 1
Telemedicine	Piloted, effective in hybrid CHW models; bandwidth-constrained	Scale near-term — priority 2
Remote Patient Monitoring & AI triage	Early pilots; depends on EHR data layer	Sequence after EHR — priority 3
Tele-rehabilitation	Almost no Kenya-specific evidence base yet	Fund as an evaluated pilot, not a scaled rollout

4. Strategic Recommendations

— Fund interoperability completion first. The Enterprise Service Bus and National Shared Registries under the Digital Health Superhighway blueprint should absorb the largest near-term share of digital health investment — every other area's value is capped by whether data can actually move between systems.

— Scale telemedicine through the CHW hybrid model, not standalone apps. The evidence favours connecting virtual consults to existing community health worker infrastructure over stand-alone consumer telehealth platforms, and edge-computing approaches should be prioritised for rural, low-bandwidth counties.

— Treat AI and remote monitoring as a data-layer investment, not a device investment. Value depends on data reaching a connected EHR — fund the connective plumbing alongside any device or algorithm procurement, and prioritise regulatory clarity on AI-in-health ahead of large-scale deployment.

— Commission a Kenya-specific tele-rehabilitation pilot with built-in evaluation. Given the country's rehabilitation workforce shortage, this is a high-potential, low-evidence area — the right first investment is a rigorously evaluated pilot, not a national rollout.

5. What This Brief Does Not Yet Cover

This is the Executive layer of Flagship Project 02. It does not include cost modelling, vendor or platform-specific evaluation, or a county-level rollout plan — those belong in this project's Analytics and Engineering layers, to be developed next.

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